

Traitement des images numériques

Morphologie mathématique

08/04/2024
Sylvain Lobry

Rappel : toute captation
(notamment audio ou vidéo) est
strictement interdite

Traitement des images

Morphologie mathématique

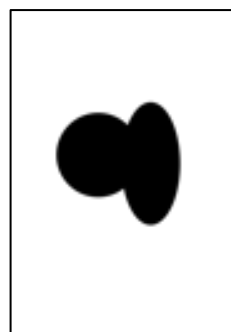
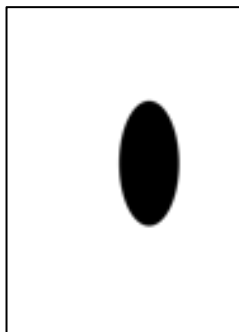
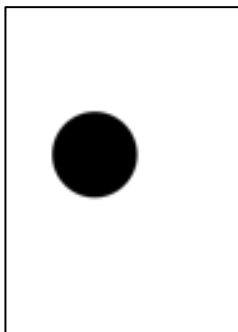
- Théorie inventée dans les années 60 par Matheron et Serra (Ecole des Mines de Paris)
- A la base, pour l'étude des roches, aujourd'hui:
 - Biologie
 - Imagerie médicale
 - Télédétection
 - Sciences des matériaux
 - ...
- Souvent présent dans une chaîne de traitement
- Théorie ensembliste

Opérations sur les ensembles

Traitement des images

Union

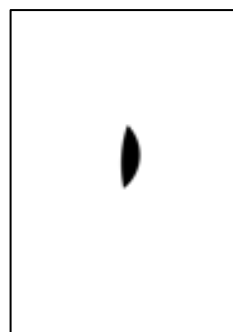
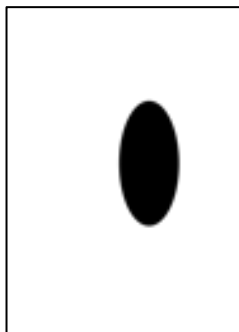
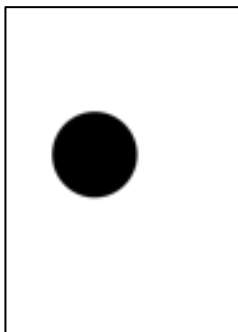
- Soit deux ensembles I et S . L'union est définie par:
$$I \cup S = \{x \mid x \in I \vee x \in S\}$$
- Exemple
- $A = \{(0, 0), (1, 1), (0, 1)\}$ et $B = \{(0, 1), (1, 0)\}$
- $A \cup B = \{(0, 0), (1, 1), (0, 1), (1, 0)\}$
- Application à des images binaires:



Traitement des images

Intesection

- Soit deux ensembles I et S . intersection est définie par:
$$I \cap S = \{x \mid x \in I \wedge x \in S\}$$
- Exemple
- $A = \{(0, 0), (1, 1), (0, 1)\}$ et $B = \{(0, 1), (1, 0)\}$
- $A \cap B = \{(0, 1)\}$
- Application à des images binaires:



Traitement des images

Somme de Minkowski

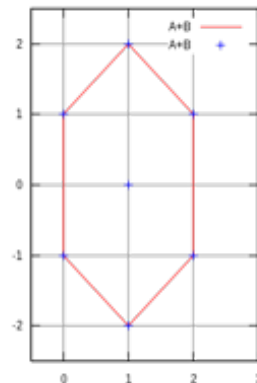
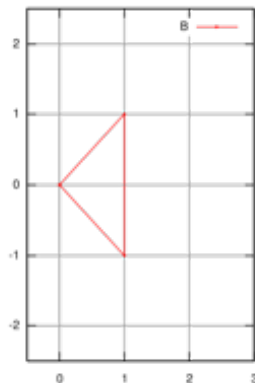
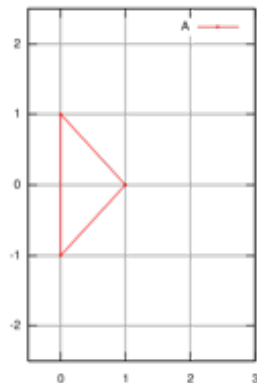
- Somme de Minkowski, définie pour deux ensembles I et S :

$$I \oplus S = \{i + s \mid i \in I, s \in S\}$$

- Exemple:

- $A = \{(0, -1), (0, 1), (1, 0)\}$ et $B = \{(0, 0), (1, -1), (1, 1)\}$

- $A \oplus B = \{(0, -1), (0, 1), (1, 0), (1, -2), (2, -1), (2, 1)\}$



Traitement des images

Somme de Minkowski

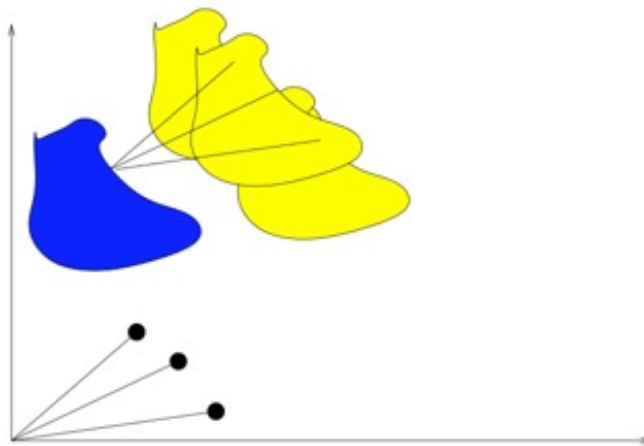
– Exemple (cours de Luc Brun)

$$X \oplus B = \bigcup_{b \in B} X + b$$

■ X : ■

■ B : ●

■ $X \oplus B$: ■

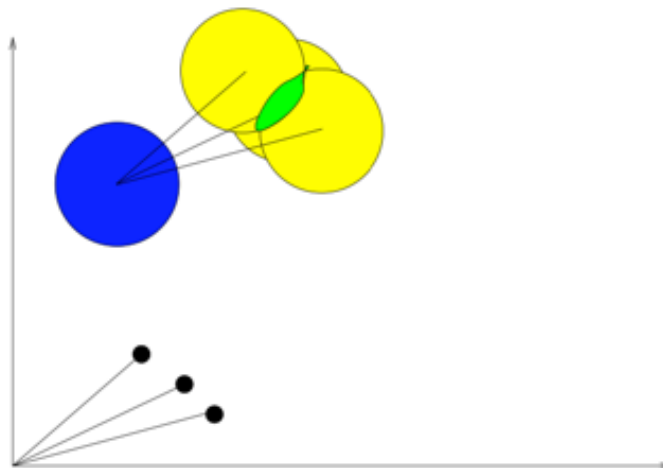


Traitement des images

Soustraction de Minkowski

$$X \ominus B = \bigcap_{b \in B} X + b$$

- X : ■
- B : ●
- $X \ominus B$: ■

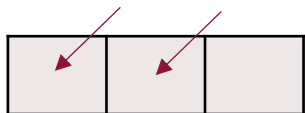


Opérations morphologique de base – cas binaire

Traitement des images

Élément structurant

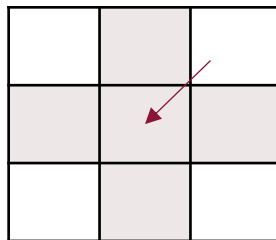
- Un élément structurant est un sous-ensemble de l'espace de l'image (dans notre cas $\mathbb{Z}^2$) centré en $(0, 0)$
- Ici, on représente l'origine $(0, 0)$ par une flèche.
- L'élément structurant S appliqué au point x est noté $S_x = \{x + s \mid s \in S\}$
- Symétrique d'un élément structurant $\check{S} = \{-s \mid s \in S\}$



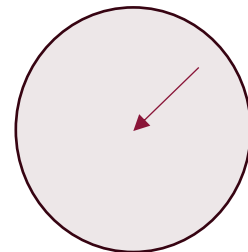
$(0, -1), (0, 0), (0, 1)$

$(0, 0), (0, 1), (0, 2)$

$(0, -3), (0, -2), (0, -1)$



$(-1, 0), (0, 1), (1, 0), (0, -1), (0, 0)$



Traitement des images

Dilatation

- La dilatation de l'image I par l'élément structurant S est la somme de Minkowski:

$$\delta_S(I) = I \oplus S = \{i + s \mid i \in I, s \in S\}$$

- Définition avec le symétrique:

$$\delta_S(I) = \{i \mid I \cap \check{S}_i \neq \emptyset\}$$

Dilatation - Example

A diagram showing a 2x2 grid of squares. A red arrow points from the top-right corner of the grid towards the center of the top-right square.

A diagram showing a 2x2 grid of squares. A red arrow points from the top-right corner of the grid towards the center of the top-left square.

0	0	1	1	0	0
0	1	1	1	0	0
0	1	1	0	1	0
1	1	1	0	0	0
0	1	1	1	0	0
0	0	1	1	1	0

[illegible]

Dilatation - Example

A diagram showing a 2x2 grid of squares. The top-left square is white, and the other three squares (top-right, bottom-left, and bottom-right) are light blue. A red arrow points from the top-right corner towards the center of the bottom-right square.

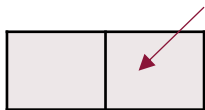
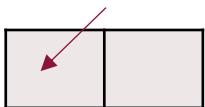
A diagram showing a 2x2 grid of squares. The top-left square is shaded light blue. A red arrow points from the top-left corner of the grid towards the center of the top-left square.

0	0	1	1	0	0
0	1	1	1	0	0
0	1	1	0	1	0
1	1	1	0	0	0
0	1	1	1	0	0
0	0	1	1	1	0

[illegible]

Traitement des images

Dilatation - Exemple

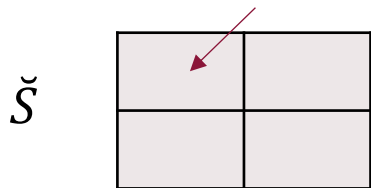
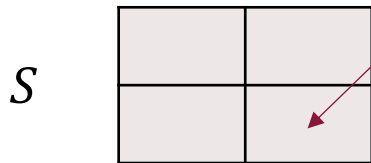
 $\mathcal{S}$

 $\check{\mathcal{S}}$


0	0	1	1	0	0
0	1	1	1	0	0
0	1	1	0	1	0
1	1	1	0	0	0
0	1	1	1	0	0
0	0	1	1	1	0

0	1	1	1	0	0
1	1	1	1	0	0
1	1	1	1	1	0
1	1	1	0	0	0
1	1	1	1	0	0
0	1	1	1	1	0

Traitement des images

Dilatation – Exemple 2



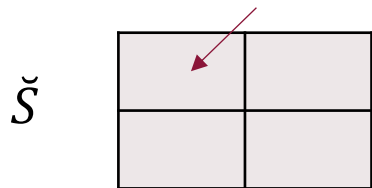
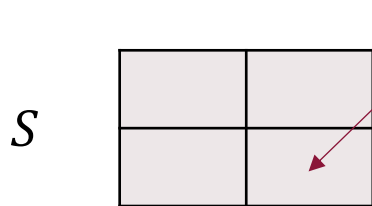
0	0	0	0	0	0
0	0	1	1	1	0
0	0	1	0	0	1
1	0	1	0	0	1
0	1	1	1	0	0
0	0	1	1	0	0

$$S = (-1, -1), (-1, 0), (0, -1), (0, 0)$$

$$\check{S} = (1, 1), (1, 0), (0, 1), (0, 0)$$

Traitement des images

Dilatation – Exemple 2

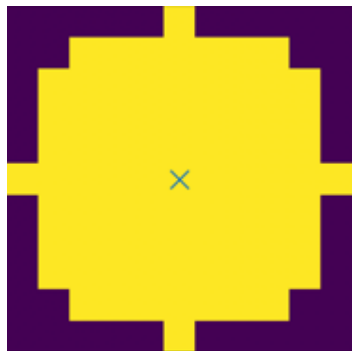


0	0	0	0	0	0
0	0	1	1	1	0
0	0	1	0	0	1
1	0	1	0	0	1
0	1	1	1	0	0
0	0	1	1	0	0

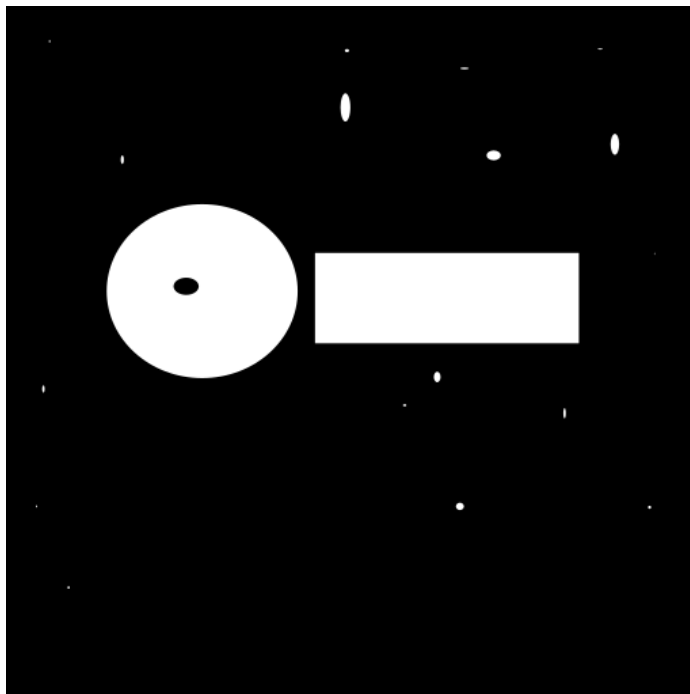
0	1	1	1	1	0
0	1	1	1	1	1
1	1	1	0	1	1
1	1	1	1	1	1
1	1	1	1	0	0
0	1	1	1	0	0

Traitement des images

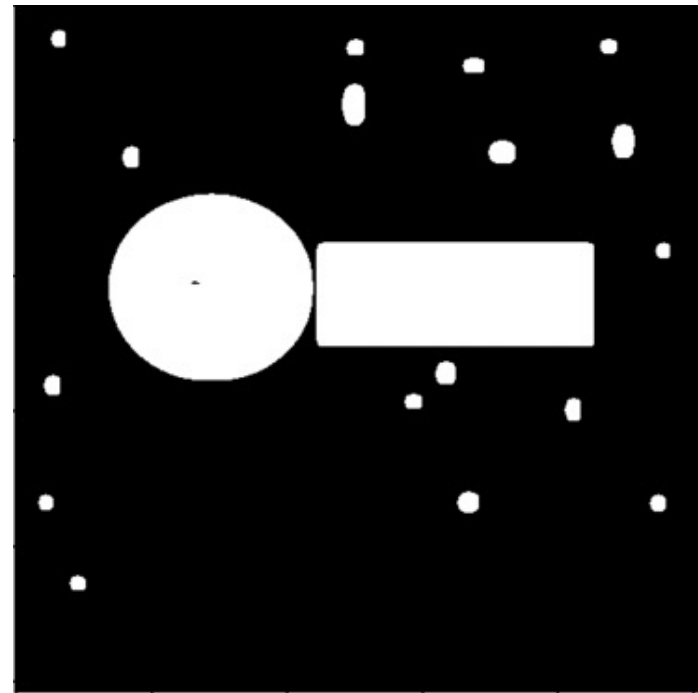
Dilatation – Exemple 3

 S


Original

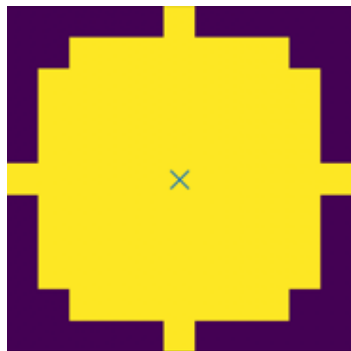


Dilaté

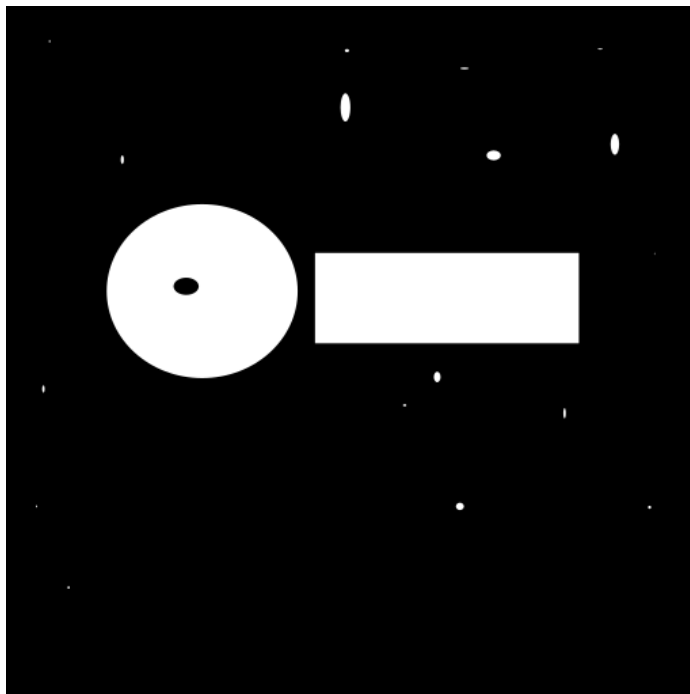


Traitement des images

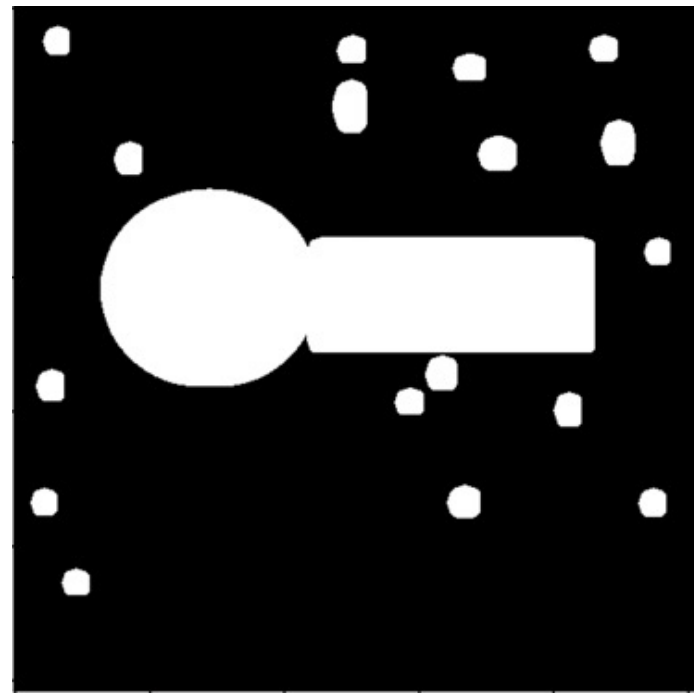
Dilatation – Exemple 3

 S


Original



Dilaté x2



Traitement des images

Effets et propriétés de la dilatation

- Bouche les trous
- Fusionne les composantes connexes proches
- Augmente la taille des formes

- Si $X \subset Y$, $\delta_S(X) \subset \delta_S(Y)$
- Si $S \subset S'$ $\delta_S(I) \subset \delta_{S'}(I)$
- $\delta_{S \oplus S'}(I) = \delta_{S'}(\delta_S(I))$

Traitement des images

Erosion

- La dilatation de l'image I par l'élément structurant S est la soustraction de Minkowski avec le symétrique:

$$\epsilon_S(I) = I \ominus \check{S} = \{i \mid \forall s \in S, i + s \in I\}$$

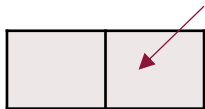
- Définition avec l'inclusion:

$$\epsilon_S(I) = \{i \mid S_i \subset I\}$$

Traitement des images

Erosion - Example

S



0	0	1	1	0	0
0	1	1	1	0	0
0	1	1	0	1	0
1	1	1	0	0	0
0	1	1	1	0	0
0	0	1	1	1	0

[illegible]

Traitement des images

Erosion - Exemple

S

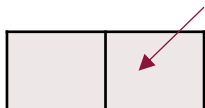
			0	0	1	1	0	0
		0	1	1	1	0	0	
		0	1	1	0	1	0	
		1	1	1	0	0	0	
		0	1	1	1	0	0	
		0	0	1	1	1	0	

0	0	0	1		

Traitement des images

Erosion - Exemple

S



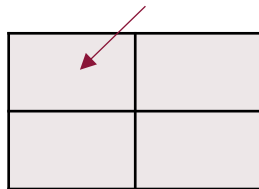
0	0	1	1	0	0
0	1	1	1	0	0
0	1	1	0	1	0
1	1	1	0	0	0
0	1	1	1	0	0
0	0	1	1	1	0

0	0	0	1	0	0
0	0	1	1	0	0
0	0	1	0	0	0
0	1	1	0	0	0
0	0	1	1	0	0
0	0	0	1	1	0

Traitement des images

Erosion – Exemple 2

S

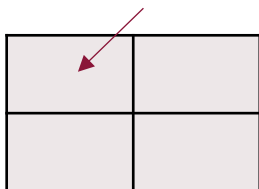


1	1	1	1	1	1
1	1	0	0	0	1
1	1	0	1	1	0
0	1	0	1	1	0
1	0	0	0	1	1
1	1	0	0	1	1

Traitement des images

Erosion – Exemple 2

S



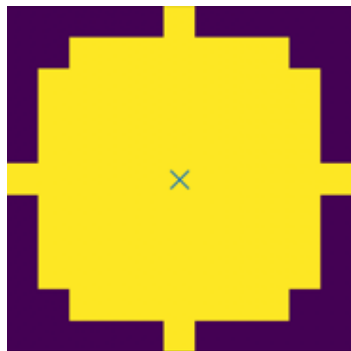
1	1	1	1	1	1
1	1	0	0	0	1
1	1	0	1	1	0
0	1	0	1	1	0
1	0	0	0	1	1
1	1	0	0	1	1

1	0	0	0	0	1
1	0	0	0	0	0
0	0	0	1	0	0
0	0	0	0	0	0
0	0	0	0	1	1
1	0	0	0	1	1

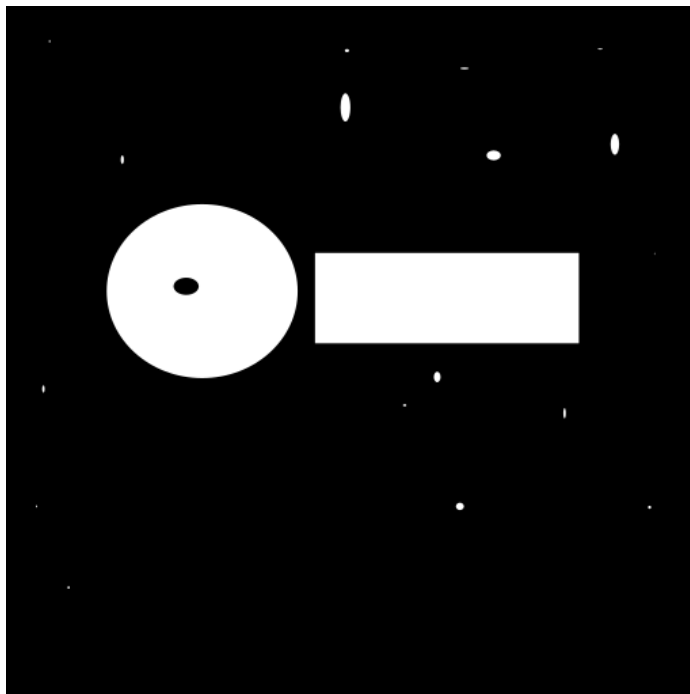
Traitement des images

Erosion – Exemple 3

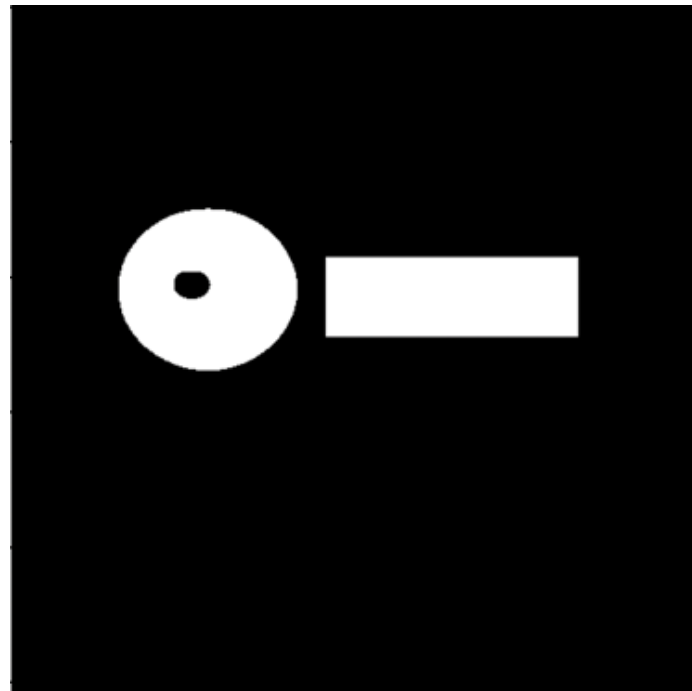
S



Original

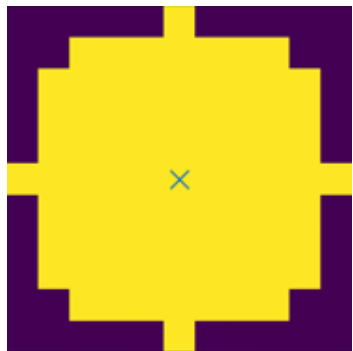


Erodé

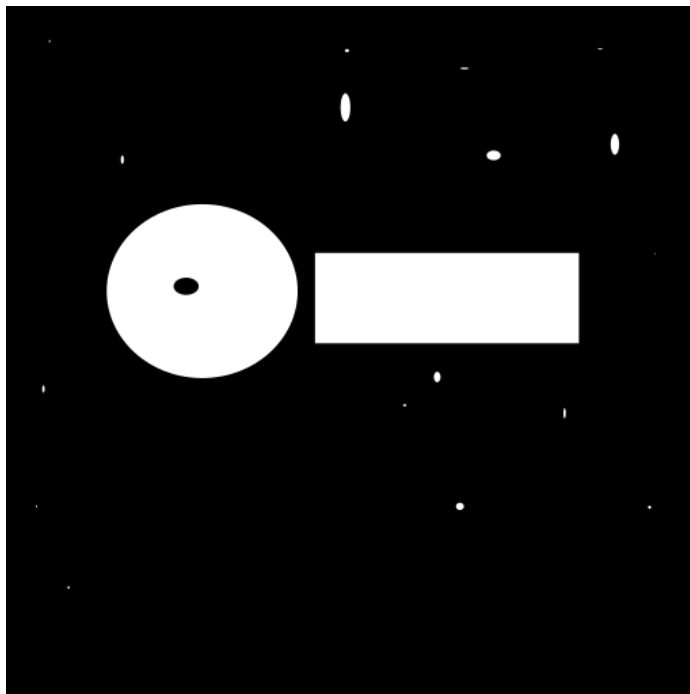


Traitement des images

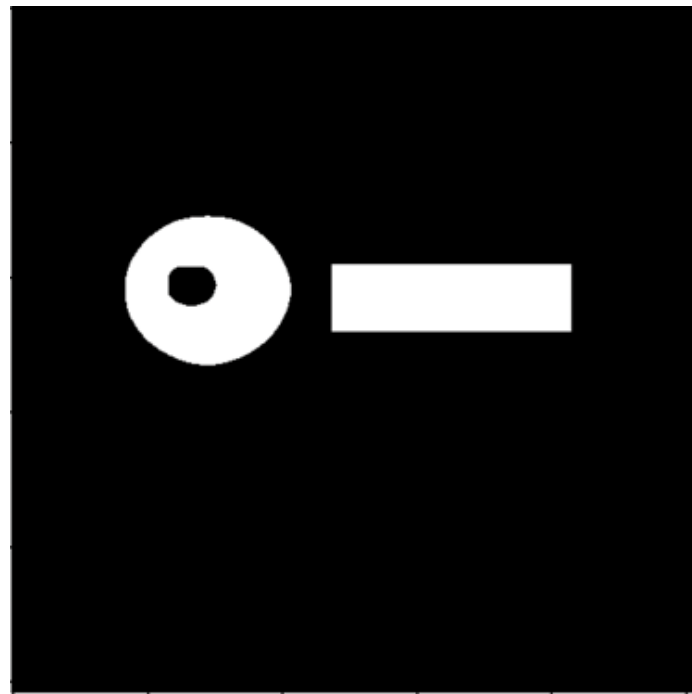
Erosion – Exemple 3

 S


Original



Erodé x2



Traitement des images

Effets et propriétés de l'érosion

- Suppression des petites composantes connexes
- Suppression des détails en bords de composante connexe
- Augmente la taille des trous
- Si $X \subset Y$, $\epsilon_S(X) \subset \epsilon_S(Y)$
- Si $S \subset S'$, $\epsilon_S(I) \subset \epsilon_{S'}(I)$
- $\epsilon_{S \oplus S'}(I) = \epsilon_{S'}(\epsilon_S(I))$